

Relation-aware Graph Transformer with Multi-view Consistency Alignment and Dynamic Fusion for Drug–Drug Interaction Prediction

Jueming Li^{1,2,3}, Shengyuan Yang^{1,2,3}, Xianfang Tang^{1,2,3}, and Qiang Zhu^{1,2,3,*}

¹Engineering Research Center of Hubei Province for Clothing Information

²Hubei Provincial Engineering Research Center for Intelligent Textile and Fashion

³School of Computer Science and Artificial Intelligence, Wuhan Textile University

*Corresponding author: qzhu@wtu.edu.cn

Abstract— Drug–Drug Interaction Prediction (DDI) is critical for ensuring safe medication. However, existing methods often rely on local structures or limited single-view molecular information, making them insufficient to capture long-range dependencies and multi-scale chemical semantics. To address these limitations, we propose MVCADF – a relation-aware Graph Transformer framework with multi-view consistency alignment and dynamic fusion for DDIs. Beyond leveraging molecular and biomedical knowledge graphs, MVCADF integrates motif fragment graphs, element-level view, and ECFP fingerprints, thus enabling comprehensive representations at the atom, fragment, element, fingerprint, and semantic levels. We design a multi-view consistency alignment strategy to project different views into a shared anchor space, which mitigates inter-view distributional shift and enhances representational coherence. This strategy is followed by a dynamic fusion module that adaptively integrates multi-view features based on their discriminative significance. Additionally, the relation- and distance-aware attention mechanism further facilitates modeling of long-range interactions within heterogeneous graphs. Extensive experiments show that MVCADF outperforms the best baseline by 3%–5% across multiple public datasets, offering a robust and holistic solution for DDIs prediction.

I. INTRODUCTION

Drug–drug interactions (DDIs) refer to the pharmacological interplay that occurs when two or more drugs are administered together, which could enhance therapeutic efficacy or diminish it [1]. DDI prediction is crucial for preventing adverse events, ensuring drug safety during development, and establishing effective monitoring protocols for combination therapies. In recent years, graph-based approaches have become a focal point for drug–drug interaction prediction. Most methods focus on two kinds of graph structures: homogeneous and heterogeneous graphs [2]. The homogeneous graphs are composed solely of drug nodes, while heterogeneous graphs are complex networks structures that contain multiple types of nodes and relationships. As a result, heterogeneous graphs offer a clear advantage over complex biological systems. There are two kinds of heterogeneous graphs: biomedical knowledge graphs (BKGs) and drug molecular graphs. BKGs represent molecular entities as nodes and their functional or regulatory relationships as edges, thereby integrating diverse

biomedical information into a unified framework. In contrast, drug molecular graphs model atoms as nodes and chemical bonds as edges, offering fine-grained representations of molecular structure. For BKGs, researchers design relation-specific transformations to encode entity semantics before aggregating information through message-passing frameworks. For drug molecular graphs, it is common to employ virtual nodes or global pooling layers to generate drug-level embeddings.

Although GNN-based techniques have advanced DDIs prediction, three key challenges remain [3]. Firstly, most current GNN methods rely solely on neighborhood aggregation, limiting their ability to capture long-range dependencies between drugs in graphs. Secondly, the distortion of information flowing from distant nodes limits the efficiency of message passing on long-distance drug–drug interactions. This phenomenon is referred to as ‘over-squashing’. Furthermore, current approaches rely on molecular-knowledge graph dual views but omit multi-level structural features. In addition, distribution misalignment across views additionally weakens synergy. To tackle these challenges, we propose MVCADF - Relation-aware Graph Transformer with Multi-view Consistency Alignment and Dynamic Fusion for Drug–Drug Interaction Prediction. Our main contributions include:

- **Multi-view consistency alignment.** Within individual views, MVCADF maximizes graph-level local mutual information to ensure the structural integrity of each view’s own structure. Across views, MVCADF employs a consistency alignment mechanism to unify all views into a coherent scale and orientation, thus resolving the view discrepancy problem and ensuring a consistent representation for subsequent view fusion.
- **Multi-view dynamic fusion.** MVCADF incorporates molecular graphs, knowledge graphs, BRICS fragment graphs, element graphs, and molecular fingerprints into a unified learning framework, and employs an adaptive multi-view approach to achieve multi-scale information fusion spanning atoms, fragments, elements, fingerprints, and semantics.

II. RELATED WORK

1. Molecular graph based methods

Molecular graphs offer an intuitive, structured depiction of a drug’s chemical architecture, laying the groundwork for uncovering potential drug–drug interactions [4]. In recent years, deep-learning models operating on these graphs—such as GCNs and GATs—have been widely adopted for DDI prediction because they can learn both atom-level local features and molecule-level global topology simultaneously [5], [6]. Recently, the Transformer-based Graphormer [7] has been introduced to the task. GNNs struggle with long-range dependencies, Graphormer directly encodes relationships between distant nodes, making it particularly effective for capturing extended interactions. [8] Subsequent work has enriched this research line by incorporating multi-scale receptive-field concatenation and learnable subgraph-pooling mechanisms. For example, GNN-DDI [9] employs a five-layer multi-head GAT combined with SAGPooling to cascade 1-5 hop substructures, improving cold-start drug generalization and enabling functional-group-level interpretability. Meanwhile, MMDDI-MGPFF [10] builds on GIN-encoded molecular graphs and additionally integrates sequence-based features—such as targets, enzymes, and pharmacological descriptors—using multi-head attention for cross-modal fusion, thereby achieving higher accuracy on event-level DDI prediction tasks. In our evaluation, SSI-DDI and Molormer serve as representative baselines for this family, assessing the effectiveness of learning drug representations directly from molecular graphs for DDI prediction.

2. Biomedical knowledge graph based methods

These approaches [11], [12] exploit the rich entity–relation semantics embedded in biomedical knowledge graphs, markedly boosting the predictive power for drug interactions. KGNN [13] was the first to introduce graph convolutional networks into this setting, using message passing to learn deep representations of drug nodes in multi-relational graphs. For multi-relation DDI tasks, KG2ECapsule [14] innovatively incorporated capsule networks for non-linear transformation, further enhancing entity expressiveness within relation-specific spaces. DDKG [15] combined graph attention networks and an attention mechanism to precisely capture key “drug–relation–drug” triple patterns, yielding more fine-grained interaction information. Building on knowledge-graph embeddings, BioDKG-DDI [16] integrates multi-scale features from molecular structure and functional similarity, employing multi-view attention to adaptively weight each feature and demonstrating stronger robustness in cross-dataset evaluations. KITE-DDI [17] feeds drug-node embeddings and randomized SMILES sequences into a lightweight Transformer; by using knowledge-graph self-attention to explicitly align “drug–drug” semantic dependencies, it significantly improves event-level prediction accuracy for cold-start drugs. In our comparative experiments, KGNN, KG2ECapsule, and DDKG are included as baselines representative of knowledge-graph-based modeling.

3. Multi-level methods

Multi-level DDI prediction approaches jointly exploit graph-level information—derived from molecular structure—and node-level information learned within drug–drug or more complex interaction networks. Although most studies still focus on homogeneous DDI graphs, several approaches have been extended to heterogeneous settings. Bi-GNN [18] constructs a two-tier graph (a top-layer drug–interaction graph and, beneath it, an unfolded molecular graph for each drug) and performs cross-layer reasoning with GAT. MIRACLE [19] encodes DDI relations using GCNs and introduces bond-aware attentive message passing within molecular graphs. MDNN [20] designs parallel pathways on a heterogeneous knowledge graph to capture topology and drug attributes simultaneously. TIGER [21] models biomedical knowledge subgraphs and drug molecular graphs in parallel. It maximizes Jensen–Shannon mutual information to align global and local representation, and employs relation-aware attention to capture long-range dependencies, achieving state-of-the-art results on multiple benchmarks. Despite validating the effectiveness of combining molecular- and network-level information, these methods still lack sufficient view scalability, cross-view consistency, and dynamic fusion strategies.

III. METHODOLOGY

The overall framework of MVCADF is depicted in the figure 1. The model simultaneously encodes five complementary views—knowledge graphs, molecular graphs, motif graphs, elemental graphs, and fingerprint vectors—and leverages multi-view dynamic fusion together with cross-view consistency alignment to derive a unified drug representation, thereby enabling more accurate prediction of potential DDIs.

A. Multi-view Drug Representation

1. Knowledge and Molecular Graphs

For knowledge graphs and molecular graphs, we use Graph-Transformer architectures based on self-attention mechanisms. Though self-attention is central to Transformers, its positional information perception fails to effectively capture the intrinsic properties of multi-type relationships. Additionally, sparse correlations between distant nodes lead to missing semantic associations, and multiple interactions often exist among the same node. To address these issues, we introduce a relationship-aware self-attention mechanism [21]. 1) Let $G = (V, E, R)$ denote an arbitrary graph, where $x_v \in \mathbb{R}^d$ represents the attribute of node $v \in V$, and all node attributes are collectively stored in a matrix $X \in \mathbb{R}^{|V| \times d}$ with dimension $|V| \times d$. For any node pair (v, u) with $c \geq 1$ types of associative relationships, we compute its attention score by integrating their intrinsic features and inter-node relational information:

$$\begin{aligned} \psi(x_v, x_u, r_{v \leftrightarrow u}^c) &= (x_v + r_{v \rightarrow u}) \mathbf{W}_q^T \mathbf{W}_k (x_u + r_{u \rightarrow v}) \\ &= x_v \mathbf{W}_q^T \mathbf{W}_k x_u + x_v \mathbf{W}_q^T \mathbf{W}_k r_{u \rightarrow v} \\ &\quad + r_{v \rightarrow u} \mathbf{W}_q^T \mathbf{W}_k x_u + r_{v \rightarrow u} \mathbf{W}_q^T \mathbf{W}_k r_{u \rightarrow v} \end{aligned} \quad (1)$$

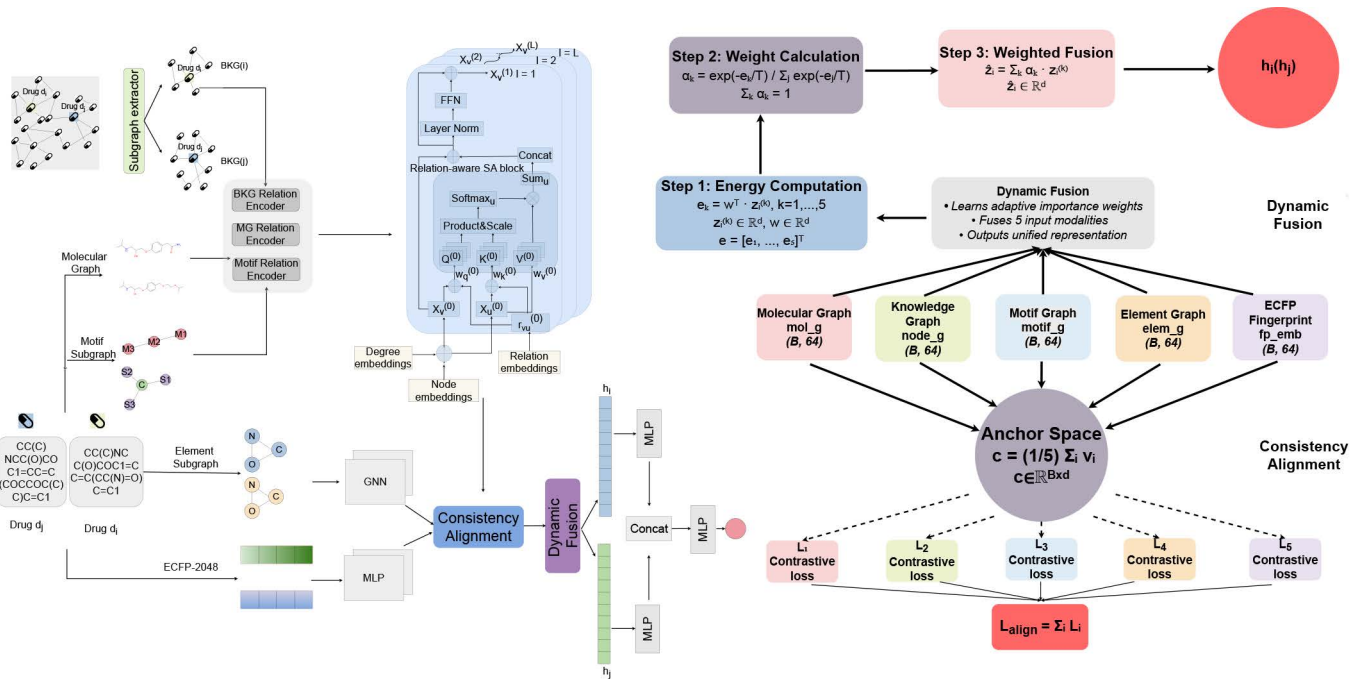


Fig. 1: Framework of the proposed MVCADF.

where $r_{v \leftrightarrow u}^c$ denotes the c -th type of associative relationship between node pair (v, u) . Specifically, when graph G is undirected, $r_{v \rightarrow u}$ and $r_{u \rightarrow v}$ take equal values. The trainable parameter matrices $\mathbf{W}_q, \mathbf{W}_k \in \mathbb{R}^{d_k \times d}$ are respectively used to generate query features Q and key features K . The newly defined function $\psi(\cdot, \cdot, \cdot)$ can incorporate both source-specific and target-specific relational information simultaneously as the model attends to different positions. Furthermore, leveraging a general relational bias term enables the model to model and reinforce shared structural patterns and dependencies across the entire input sequence. Notably, for long-range node pairs, this term can be initialized based on the shortest path between nodes or efficiently extracts structural properties between them. Based on this, the relationship-aware self-attention mechanism is as follows:

$$\text{RAttn}(x_v) = \sum_{u \in V} \frac{\sum_{c \in C} f(x_v, x_u, r_{v \leftrightarrow u}^c)}{\sum_{w \in V} \sum_{c \in C} f(x_v, x_w, r_{v \leftrightarrow w}^c)} \phi(x_u) \quad (2)$$

where $f(\cdot, \cdot, \cdot) = \exp(\psi(\cdot, \cdot, \cdot) / \sqrt{d_k})$, $\phi(x_u) = \mathbf{W}_v x_u$. $\mathbf{W}_v \in \mathbb{R}^{d_v \times d}$ is a learnable parameter to get the representation V .

2) To enhance the expressive capability of graph structural information, we further integrate position embeddings designed based on node degrees into node features, with the specific implementation as follows:

$$x_v^{(0)} = x_v + z_{deg(v)}, \forall v \in V \quad (3)$$

where $x_v^{(0)}$ is the input of the relationship-aware self-attention module. $deg(v)$ is the degree of node v , and $z(\cdot)$ is a parameterized embedding layer designed based on node degrees.

3) The output $\text{RAttn}(x_v^{(0)})$ from the relationship-aware self-attention module undergoes processing via a skip connection

and a two-layer feed-forward neural network (FFN), ultimately yielding the updated node representation $x_v^{(1)}$. The formal definition is as follows:

$$\hat{\mathbf{X}}^{(\ell)} = \mathbf{X}^{(\ell-1)} + \text{RAttn}(\mathbf{X}^{(\ell-1)}) \quad (4)$$

$$\mathbf{X}^{(\ell)} = \text{FFN}(\hat{\mathbf{X}}^{(\ell)}) = \text{ReLU}(\hat{\mathbf{X}}^{(\ell)} \mathbf{W}_1^{(\ell)}) \mathbf{W}_2^{(\ell)} \quad (5)$$

where $\mathbf{X}^{(\ell)}$ denotes the updated representation. $\mathbf{W}_1^{(\ell)}$ and $\mathbf{W}_2^{(\ell)}$ are the trainable parameters of ℓ -th layer ($1 \leq \ell \leq L$).

2. Motif Graph

Motif graphs first decompose molecules into P functional group fragments via chemical rules. Since the node size of each fragment is significantly smaller than that of the atom graph, we simplify the depth of the Graph-Transformer from 3 layers to 2 layers, enabling lower computational cost and stronger interpretability when focusing on key substructures. The specific process is as follows: First, we retain only two statistical features—"number of atoms" and "number of bonds"—for each fragment. These two scalars are then concatenated into a matrix $\mathbf{X} \in \mathbb{R}^{P \times 2}$ and mapped to the model's internal d -dimensional space as the initial node representations.

$$\mathbf{H}^{(0)} = \mathbf{X} \mathbf{W}_e + b_e \quad (6)$$

where $\mathbf{W}_e \in \mathbb{R}^{2 \times d}$ and $b_e \in \mathbb{R}^d$. These nodes and edges formed by shared atoms (with each edge augmented by a shortest-path distance bias $b(d_{ij})$) are then fed into a two-layer relational-aware multi-head self-attention network. At the ℓ -th layer, the attention energy for each directed edge is:

$$e_{ij} = \frac{(q_i + r_{ij})^\top (k_j + r_{ij})}{\sqrt{d}} + b(d_{ij}) \quad (7)$$

where $q_i = W_q H_i^{(\ell)}$ and $k_j = W_k H_j^{(\ell)}$ denote the query and key vectors, respectively. r_{ij} is the relational embedding. After normalizing the attention weights, their aggregation yields the update $M^{(\ell)}$, which is then used to generate the next-layer representations via residual connections and a lightweight feed-forward network.

Finally, global average pooling is applied to the final outputs $H^{(2)}$ of all P fragment nodes to generate a compact vector:

$$g_{\text{motif}} = \frac{1}{P} \sum_{i=1}^P H_i^{(2)} \in \mathbb{R}^d \quad (8)$$

This approach captures functional group distribution and long-range fragment connectivity. The Motif channel reduces computational costs, preserves atomic graph granularity, and supplements functional group-level semantics with mesoscopic inter-fragment dependencies.

3. Element Graph

In this view, the element graph channel focuses on prior information at the level of "which chemical elements are present in the molecule and how they interact". Its operational process and corresponding formulas are as follows:

$$\tilde{A} = D^{-\frac{1}{2}} (A + I) D^{-\frac{1}{2}} \quad (9)$$

$$D_{ii} = \sum_j (A_{ij} + I_{ij}) \quad (10)$$

where $A \in \mathbb{R}^{N \times N}$ denote the adjacency matrix. I denotes the identity matrix. Next, initial feature matrix $X \in \mathbb{R}^{N \times d_0}$ is built from element properties (e.g., periodic position, electron configuration). The first GCN layer linearly fuses neighbor co-occurrence signals with these features, then applies a nonlinear activation:

$$H^{(1)} = \text{ReLU}(\tilde{A} X W^{(0)}) \quad (11)$$

where $W^{(0)} \in \mathbb{R}^{d_0 \times d}$ is the first layer's trainable weight matrix (output dim d matches other channels). The second GCN layer re-aggregates activated features to capture higher-order cross-element co-occurrence:

$$H^{(2)} = \text{ReLU}(\tilde{A} H^{(1)} W^{(1)}) \quad (12)$$

where $W^{(1)} \in \mathbb{R}^{d \times d}$ is the weight matrix. we apply global average pooling to the outputs of all element nodes to obtain the graph-level representation of the element channel:

$$h_{\text{elem}} = \frac{1}{N} \sum_{i=1}^N H_{i,:}^{(2)} \in \mathbb{R}^d \quad (13)$$

Through the above two-layer GCN and pooling operations, the element graph efficiently captures the periodic table's horizontal (same-period) and vertical (same-group) co-occurrence priors, complements structural/fragment views, and enhances drug chemical semantics modeling.

4. Molecular Fingerprint

The molecular fingerprint, treated as a generic feature vector, is compressed into a d -dimensional latent space consistent with the graph channel via a two-layer fully connected network:

$$h^{(1)} = \text{Dropout}(\text{ReLU}(x_{\text{fp}} W_1 + b_1)) \quad (14)$$

where $W_1 \in \mathbb{R}^{2048 \times d'}$. The first layer maps the input 2048-dimensional binary fingerprint to an intermediate dimension, incorporates ReLU and Dropout, which not only compresses information but also introduces regularization effects;

$$h_{\text{fp}} = h^{(1)} W_2 + b_2 \quad (15)$$

where $W_2 \in \mathbb{R}^{d' \times d}$. The second layer transforms the intermediate representation into a d -dimensional output via linear projection, maintaining consistent dimensionality with channels such as the molecular graph. This design preserves the fingerprint's efficient encoding capability for local substructures (e.g., rings, functional groups), integrates into end-to-end deep learning frameworks, complements the global priors of graph structure channels, and enhances the completeness and discriminative power of the fused representation.

B. Multi-View Consistency Alignment

In drug-drug interaction (DDI) scenarios, multi-view representations can systematically alleviate the limitations of any single view; however, prior work has shown that when views differ in semantics or quality, naively performing multi-view contrast can induce *representation degeneration* of high-quality views and weaken overall separability. It is therefore necessary to conduct **cross-view consistency and scale calibration** before fusion [22]. Meanwhile, DDI information is inherently *multi-dimensional and multi-granular* (e.g., 2D molecular graphs, motif graphs, 3D structures, network neighborhoods, and 1D fingerprints). Simple concatenation or static weighting often fails to fully exploit complementary signals across heterogeneous views while maintaining robust decision boundaries—an observation repeatedly emphasized by recent multi-dimensional and multi-view DDI methods [23], [24]. Moreover, multi-view contrast/balancing has been validated as an effective route to improve DDI representation learning [19]. Event-centric DDI further exhibits *data sparsity and class imbalance*, calling for contrastive mechanisms that preserve clear boundaries for low-frequency samples [25].

To this end, we introduce a **prototype-based multi-view alignment** module as a distribution calibration layer *prior* to Dynamic Fusion. Given the five view-specific embeddings of drug d_i , $\{z_i^{(k)}\}_{k=1}^5 \in \mathbb{R}^d$, we first perform ℓ_2 normalization to project each onto the unit hypersphere:

$$\hat{z}_i^{(k)} = \frac{z_i^{(k)}}{\|z_i^{(k)}\|_2}, \quad k = 1, \dots, 5. \quad (16)$$

We then apply an "equal-vote" averaging to obtain a **shared anchor** (prototype), followed by normalization:

$$\bar{z}_i = \frac{1}{5} \sum_{k=1}^5 \hat{z}_i^{(k)}, \quad \hat{\bar{z}}_i = \frac{\bar{z}_i}{\|\bar{z}_i\|_2} \quad (17)$$

During training, $(\hat{z}_i^{(k)}, \hat{z}_i)$ serves as the **positive pair**, whereas $(\hat{z}_i^{(k)}, \hat{z}_{\pi(i)})$ constructed by mini-batch shuffling of anchors acts as the **negative pair**. With temperature τ , we obtain the scaled inner products

$$s_i^{(k)} = \frac{\langle \hat{z}_i^{(k)}, \hat{z}_i \rangle}{\tau}, \quad \tilde{s}_i^{(k)} = \frac{\langle \hat{z}_i^{(k)}, \hat{z}_{\pi(i)} \rangle}{\tau} \quad (18)$$

forming an InfoNCE-style *view-to-prototype* objective. Unlike pairwise view-to-view contrast, this **prototype alignment** more effectively *suppresses the drag of low-quality views and accentuates the contribution of high-quality views*, thereby mitigating representation degeneration and improving downstream performance [22]. In addition, batch-wise shuffled anchors naturally provide abundant negatives, which helps maintain clear boundaries under *sparsity and imbalance* (consistent with observations in event-centric DDI) [25].

From the optimization perspective, the shared anchor $\bar{z}_i$ is the mean of five views; its gradient backpropagates to all view encoders simultaneously, pulling each $z_i^{(k)}$ towards $\bar{z}_i$ during training. Consequently, three key effects are achieved: (i) the scale and direction of all $z_i^{(k)}$ are aligned first, effectively preventing an “outlier” view from receiving extreme weights in dynamic fusion; (ii) within this unified coordinate system, complementary information aggregates more effectively, while noise is more easily cancels out, yielding a synergistic effect of denoising and complementation; and (iii) the “align-then-fuse” strategy conforms to the desiderata of *adaptive/attention-based fusion* in multi-dimensional DDI [19], [23], [24].

C. Dynamic Fusion

In multi-view representation learning, views have complementary yet redundant info, with varying importance per drug sample. To address this, models must go beyond fixed averaging or concatenation. Post multi-view consistency alignment (same scale and direction), we use the Multi-View Dynamic Fusion module, which adaptively and differentially assigns view weights per sample. Its core fuses vectors from parallel encoding of 5 modalities (molecular graph, KG subgraph, Motif, element, fingerprint) into a unified drug representation. First, calculate the energy score. For $\{z^{(k)}\}_{k=1}^5$, where k is the index of different views, shared linear projection computes energy scores:

$$e_k = \mathbf{w}^\top \mathbf{z}^{(k)} \quad (19)$$

where $\mathbf{w} \in \mathbb{R}^d$ is a learnable parameter, and $\mathbf{z}^{(k)}$ has been unified to the same dimension d via the preceding encoding network. This energy reflects the model’s confidence in the view for the sample: lower values mean more reliable representations. Next, temperature-scaled Softmax converts scalar energy to non-negative weights summing to 1, using $T > 0$:

$$\alpha_k = \frac{\exp(-e_k/T)}{\sum_{j=1}^5 \exp(-e_j/T)} \quad (20)$$

where the negative sign ensures that smaller energy values correspond to larger weights, aligning with the intuition that “low energy equals a good view.” The temperature T controls

the smoothness of the weight distribution: a smaller T makes the Softmax produce more extreme allocations, while a larger T leads to more evenly distributed weights across all views.

Finally, the original view representations are linearly combined using the resulting weights $\{\alpha_k\}$:

$$\tilde{\mathbf{z}} = \sum_{k=1}^5 \alpha_k \mathbf{z}^{(k)} \quad (21)$$

where $\tilde{\mathbf{z}} \in \mathbb{R}^d$ represents the drug’s unified features, integrating complementary view signals and downweighting noise or irrelevant views. This “energy-normalization-fusion” way preserves multi-view complementarity, enables fine-grained view importance regulation with smaller cost, and supports downstream tasks.

After the Dynamic Fusion generates each drug’s fused vector $\tilde{\mathbf{z}} \in \mathbb{R}^d$, the next step combines two drugs’ features into an interaction representation for classification. First, a two-layer multilayer perceptron reprocesses single drug $\tilde{\mathbf{z}}$ to project it from d -dim to a more discriminative interaction subspace, outputting $\mathbf{h}_i, \mathbf{h}_j \in \mathbb{R}^d$. This projection layer (linear mapping, ReLU, Dropout) enhances feature separability for sample discrimination and aligns multi-task representation spaces via lightweight ops. Finally, the Hadamard product of $\mathbf{h}_i$ and $\mathbf{h}_j$ yields the interaction vector.

$$\mathbf{g}_{ij} = \mathbf{h}_i \odot \mathbf{h}_j \quad (22)$$

It retains feature-wise similarity info and symmetry, naturally suiting unordered drug pair interaction modeling. Then, concatenate $\mathbf{h}_i$, $\mathbf{h}_j$ and $\mathbf{g}_{ij}$ into a $3d$ -dimensional interaction feature.

$$\mathbf{u}_{ij} = [\mathbf{h}_i \parallel \mathbf{h}_j \parallel \mathbf{g}_{ij}] \quad (23)$$

Next, $\mathbf{u}_{ij}$ is fed into a 3-layer fully connected classifier ($3d \rightarrow 256 \rightarrow 512 \rightarrow 2$). After ReLU and Dropout, logits are output and softmaxed to get interaction probability P_{ij} . This design enhances discriminativity via projections, captures symmetric patterns via Hadamard product, and models nonlinearity via the deep classifier. Ultimately, it compresses multi-view fused high-dim info into an “interaction probability” for end-to-end DDI prediction, balancing regularization stability and scalability.

D. Loss Function

While maintaining the end-to-end DDI link prediction, we incorporate two layers of contrastive consistency losses: one is the intra-channel global-local InfoMax contrastive loss, and the other is the multi-view consistency alignment loss.

1. Global-local InfoMax contrastive loss

For each drug d_i , the model uses multi-view encoding, dynamic fusion, and projection to get a global representation $\mathbf{h}_i \in \mathbb{R}^d$. Meanwhile, molecular graph channel Graph-Transformer layer embeddings form local $\{\mathbf{s}_i^{\text{mol}}\}$; knowledge subgraph channel yields $\{\mathbf{s}_i^{\text{bkg}}\}$. To maintain high consistency between the global vector $\mathbf{h}_i$ and the drug’s local vectors, we construct positive and negative samples as follows: The positive pairs $(\mathbf{h}_i, \mathbf{s}_i)$ are from the same drug; For negative

pairs, we first shuffle $\mathbf{h}_i$ within the batch to generate $\tilde{\mathbf{h}}_i$, and then form pairs as $(\tilde{\mathbf{h}}_i, \mathbf{s}_i)$. A bilinear discriminator scores similarity logits for these pairs.

$$D(\mathbf{u}, \mathbf{v}) = \mathbf{u}^\top \mathbf{W} \mathbf{v} + b \quad (24)$$

Finally, we compute the contrastive loss with binary cross-entropy, using a *Sigmoid* activation to pull positive pairs closer and push negative pairs further apart, expressed as:

$$L_{\text{molMI}} = \frac{1}{N} \sum_{i=1}^N \left[-\log \sigma(D(h_i, s_i^{\text{mol}})) - \log(1 - \sigma(D(\tilde{h}_i, s_i^{\text{mol}}))) \right] \quad (25)$$

For the knowledge-graph L_{bkgMI} , its computation follows a similar process. Through the InfoMax constraint, the fused global vector maintains informational consistency with its corresponding local substructures without relying on classification labels, thereby simultaneously condensing both the macro-semantic and micro-chemical details of drugs within a single representation and enabling complementarity between dual-scale information.

2. Multi-view Consistency Alignment Loss

For each view k , we compute a cross-entropy style prototype-contrastive loss:

$$L_{\text{align}}^{(k)} = -\frac{1}{B} \sum_{i=1}^B \left[\log \sigma(s_i^{(k)}) + \log(1 - \sigma(\tilde{s}_i^{(k)})) \right] \quad (26)$$

and sum over the five views to obtain

$$L_{\text{align}} = \sum_{k=1}^5 L_{\text{align}}^{(k)} \quad (27)$$

This objective adheres to a *view-to-prototype* contrastive paradigm, which effectively mitigates *representation degeneration* in multi-view contrast and yields stable decision boundaries under class-imbalance settings [22], [25]. In conjunction with InfoMax in an orthogonal manner, the model enforces *cross-view consistency prior* to fusion, and—*post* fusion—imposes a *local-global* constraint to ensure the global embedding carries interpretable, structure-level evidence [19], [23], [24]. The overall training objective is

$$L = L_{\text{label}} + \alpha L_{\text{molMI}} + \beta L_{\text{bkgMI}} + \gamma L_{\text{align}} \quad (28)$$

where α , β , and γ are hyperparameters. Jointly minimizing these four terms in an end-to-end manner equips the model with multi-view-consistent drug representations, thereby enhancing the robustness and accuracy of DDI prediction.

IV. EXPERIMENT

1. Datasets

MVCADF is evaluated on two benchmark drug-related graph datasets.

- **DrugBank:** We extract verified drug–drug interaction information from the official DrugBank files. By compiling pairs of drug identifiers, we build an interaction-edge list containing 10,630 approved DDIs that span 1,052 drugs.

- **KEGG:** We parse DDI data from the KEGG database and map the resulting records to DrugBank identifiers, yielding a dataset with 13,787 approved DDIs covering 786 approved drugs.

For DrugBank and KEGG drug datasets, we collect the SMILES representation for each drug, convert to 2D molecular graphs via RDKit. For unparsable SMILES, we remove the drug and its interaction samples during preprocessing to ensure molecular view integrity. Based on molecular graphs, we extract multimodal features (BRICS Motif, ECFP-2048 fingerprints, elemental subgraphs) for MVCADF’s multi-view encoding. To avoid data leakage, Bio-KG construction excludes explicit DDI edges: we retain only heterogeneous links between drugs and proteins/biological pathways/adverse reactions, excluding drug-drug interactions, ensuring non-interaction structural info only.

2. Experimental results

The systematic comparison between MVCADF and other models on DrugBank and KEGG datasets is shown in Table 1. MVCADF outperforms across all four evaluation metrics in the DrugBank dataset, achieving 0.8300 (ACC), 0.8382 (F1), 0.9035 (AUC), and 0.8900 (AUPR)—with average gains of 3.95, 3.49, 3.73, and 5.30 over the previous best baseline. In the KEGG dataset, it achieves 0.9004 (ACC), 0.9030 (F1), 0.9567 (AUC), and 0.9447 (AUPR). Further analysis reveals that in highly sparse and noisy settings such as DrugBank, the multi-view holistic representation and Dynamic Fusion’s adaptive weighting are pivotal, confirming that the consistency alignment mechanism plus multi-view dynamic fusion substantially enhance robustness and discriminative power. In sum, MVCADF delivers stable, significant advantages across sparse vs. dense and homogeneous vs. heterogeneous data environments, demonstrating its effectiveness and versatility as a next-generation framework for DDI prediction.

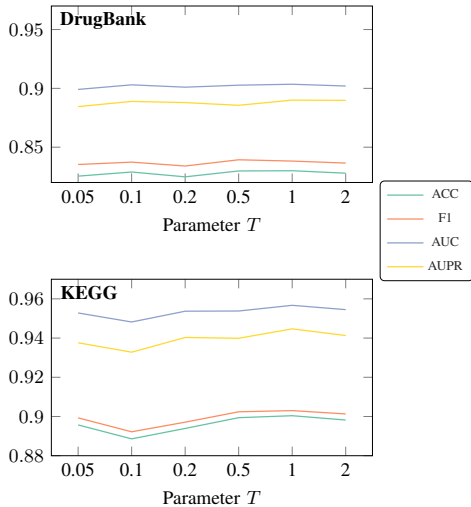
3. Ablation study

The ablation results show that MVCADF’s three key components—consistency alignment and multi-view dynamic fusion—provide progressive gains. On the DrugBank dataset, simply adding the fragment graph, element graph, fingerprint view, and dynamic soft fusion already lifts the AUC from 0.86 to 0.8701. Introducing only the consistency alignment loss (align) further raises the AUC to 0.8807 and the AUPR to 0.8621, indicating that cross-view consistency is especially critical in sparse settings. When all three components are present, AUC and AUPR jump to 0.9035 and 0.8900, respectively—an extra 2.6/3.2% over Align-LossOnly—demonstrating that sample-level weight allocation fully unleashes the discriminative power of the aligned information. The KEGG dataset exhibits the same trend: the complete model pushes AUC and AUPR to 0.9567 and 0.9447, respectively. These findings confirm that multi-view dynamic fusion and consistency alignment are both indispensable, together enabling MVCADF’s all-around superiority in both sparse and dense scenarios.

4. Parameter analysis

TABLE I: Comparison of different methods on DrugBank and KEGG datasets

Method	DrugBank				KEGG			
	ACC	F1	AUC	AUPR	ACC	F1	AUC	AUPR
SSI-DDI	0.6847 (± 0.71)	0.6830 (± 1.77)	0.7510 (± 1.12)	0.7316 (± 1.42)	0.7722 (± 1.32)	0.7795 (± 1.47)	0.8506 (± 1.41)	0.8293 (± 1.71)
Molormer	0.5763 (± 1.12)	0.5941 (± 1.13)	0.6086 (± 1.05)	0.5990 (± 0.87)	0.6371 (± 2.72)	0.6608 (± 1.54)	0.6916 (± 3.69)	0.6760 (± 3.88)
KGNN	0.6681 (± 0.56)	0.6784 (± 1.96)	0.7219 (± 0.56)	0.6614 (± 0.71)	0.7802 (± 0.95)	0.7815 (± 1.14)	0.8536 (± 0.54)	0.8185 (± 0.79)
KG2ECapsule	0.6628 (± 0.47)	0.6694 (± 0.42)	0.7131 (± 0.63)	0.6557 (± 0.92)	0.8227 (± 0.55)	0.8278 (± 0.49)	0.8960 (± 0.44)	0.8599 (± 0.69)
DDKG	0.7035 (± 1.04)	0.7089 (± 2.53)	0.7638 (± 0.32)	0.7292 (± 0.50)	0.8305 (± 0.83)	0.8355 (± 0.85)	0.9000 (± 0.75)	0.8690 (± 1.18)
Bi-GNN	0.7165 (± 2.18)	0.7419 (± 0.25)	0.7672 (± 1.11)	0.7454 (± 0.84)	0.8503 (± 0.60)	0.8580 (± 0.58)	0.9147 (± 0.63)	0.8840 (± 0.95)
MIRACLE	0.6636 (± 0.58)	0.6630 (± 0.41)	0.7102 (± 0.74)	0.6698 (± 0.86)	0.8379 (± 0.42)	0.8421 (± 0.81)	0.9090 (± 0.25)	0.8801 (± 0.53)
MDNN	0.7394 (± 0.21)	0.7313 (± 0.43)	0.8052 (± 0.36)	0.7653 (± 0.98)	0.8410 (± 0.42)	0.8459 (± 0.28)	0.9099 (± 0.41)	0.8821 (± 0.56)
TIGER-P	0.7905 (± 0.87)	0.8033 (± 0.94)	0.8662 (± 0.57)	0.8370 (± 0.68)	0.8850 (± 0.19)	0.8899 (± 0.24)	0.9473 (± 0.21)	0.9352 (± 0.35)
MVCADF (view)	0.8031 (± 0.43)	0.8179 (± 0.57)	0.8701 (± 0.54)	0.8404 (± 0.71)	0.8924 (± 0.41)	0.8957 (± 0.39)	0.9495 (± 0.32)	0.9339 (± 0.51)
MVCADF (align)	0.8060 (± 0.94)	0.8184 (± 0.75)	0.8807 (± 1.15)	0.8621 (± 1.55)	0.8918 (± 0.33)	0.8958 (± 0.35)	0.9509 (± 0.33)	0.9379 (± 0.44)
MVCADF (view&align)	0.8300 (± 0.23)	0.8382 (± 0.18)	0.9035 (± 0.39)	0.8900 (± 0.73)	0.9004 (± 0.36)	0.9030 (± 0.38)	0.9567 (± 0.19)	0.9447 (± 0.20)
Improv. (%)	3.95	3.49	3.73	5.30	1.54	1.31	0.94	0.95

Fig. 2: Different temperature parameter T of Dynamic Fusion on the DrugBank and KEGG datasets.

As illustrated in Fig 2, when the temperature parameter T is set to $\{0.05, 0.1, 0.2, 0.5, 1, 2\}$, the four evaluation metrics of Dynamic Fusion—ACC, F1, AUC, and AUPR—rise steadily on both the DrugBank and KEGG datasets as T increases from 0.05 to 1. DrugBank reaches its peak at $T = 1$ (ACC=0.8300, F1=0.8382, AUC=0.9035, AUPR=0.8900), whereas KEGG attains its best performance between $T = 1$ and $T = 1.5$ (ACC=0.9004, F1=0.9030, AUC=0.9567, AUPR=0.9447). When T is further raised to 2, all metrics decline slightly, indicating that an overly high temperature over-smooths the soft weights and dampens inter-view diversity. Conversely, when $T \leq 0.1$, the weight distribution becomes overly peaked,

causing the model to favour a single view and degrade overall performance. Notably, within the relatively broad interval $T \in [0.1, 2]$, the four metrics fluctuate only within ± 0.006 on DrugBank and ± 0.01 on KEGG, demonstrating Dynamic Fusion’s high tolerance and robustness with respect to the temperature hyper-parameter. Balancing peak performance and stability, all subsequent experiments fix the temperature parameter at $T = 1$.

V. CONCLUSION

This paper introduces a multiple view relation-aware Graph-Transformer framework for drug–drug interaction prediction named MVCADF. Our proposed MVCADF framework moves beyond the traditional molecular graph and biomedical knowledge graph approach by constructing a unified, multi-level drug representation from three novel views: BRICS fragment graphs, elemental co-occurrence graphs, and ECFP fingerprints. The model employs a relation-aware Graph Transformer to capture long-range dependences, followed by a dynamic fusion module that learns to prioritize the most discriminative views for each sample adaptively. To address cross-view distribution drift, a multi-view consistency alignment loss pulls all view embeddings toward a shared anchor before fusion, reinforcing chemical-semantic coherence. Experiments demonstrate that MVCADF achieves state-of-the-art performance on the DrugBank and KEGG benchmarks. These results confirm the synergistic advantages of multi-view information complementation, multi-view consistency alignment regularisation, and dynamic soft fusion, providing richer structural and chemical cues for deciphering complex pharmacological interaction mechanisms.

VI. ACKNOWLEDGMENT

This work is supported by the Ministry of Education Humanities and Social Sciences General project and the Hubei Provincial Natural Science Foundation of China (2024AFB736).

REFERENCES

- [1] Q. Ye, C.-Y. Hsieh, Z. Yang, Y. Kang, J. Chen, D. Cao, S. He, and T. Hou, "A unified drug-target interaction prediction framework based on knowledge graph and recommendation system," *Nature communications*, vol. 12, no. 1, p. 6775, 2021.
- [2] F. Tanvir, K. M. Saifuddin, M. Ifte Khairul Islam, and E. Akbas, "Predicting drug-drug interactions using heterogeneous graph attention networks," in *Proceedings of the 14th ACM International Conference on Bioinformatics, Computational Biology, and Health Informatics*, pp. 1–6, 2023.
- [3] D. Chen, L. O'Bray, and K. Borgwardt, "Structure-aware transformer for graph representation learning," in *International conference on machine learning*, pp. 3469–3489, PMLR, 2022.
- [4] S. Vilar, E. Uriarte, L. Santana, T. Lorberbaum, G. Hripsak, C. Friedman, and N. P. Tatonetti, "Similarity-based modeling in large-scale prediction of drug-drug interactions," *Nature protocols*, vol. 9, no. 9, pp. 2147–2163, 2014.
- [5] Z. Guo, K. Guo, B. Nan, Y. Tian, R. G. Iyer, Y. Ma, O. Wiest, X. Zhang, W. Wang, C. Zhang, and N. V. Chawla, "Graph-based molecular representation learning," in *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI-23* (E. Elkind, ed.), pp. 6638–6646, International Joint Conferences on Artificial Intelligence Organization, 8 2023. Survey Track.
- [6] A. K. Nyamabo, H. Yu, and J.-Y. Shi, "Ssi-ddi: substructure-substructure interactions for drug-drug interaction prediction," *Briefings in Bioinformatics*, vol. 22, no. 6, p. bbab133, 2021.
- [7] C. Ying, T. Cai, S. Luo, S. Zheng, G. Ke, D. He, Y. Shen, and T.-Y. Liu, "Do transformers really perform badly for graph representation?," *Advances in neural information processing systems*, vol. 34, pp. 28877–28888, 2021.
- [8] X. Zhang, G. Wang, X. Meng, S. Wang, Y. Zhang, A. Rodriguez-Paton, J. Wang, and X. Wang, "Molormer: a lightweight self-attention-based method focused on spatial structure of molecular graph for drug-drug interactions prediction," *Briefings in Bioinformatics*, vol. 23, no. 5, 2022.
- [9] Y.-H. Feng and S.-W. Zhang, "Prediction of drug-drug interaction using an attention-based graph neural network on drug molecular graphs," *Molecules*, vol. 27, no. 9, p. 3004, 2022.
- [10] G. Wang, Z. Guo, G. You, M. Xu, C. Cao, and X. Hu, "Mmddi-mgpf: Multi-modal drug representation learning with molecular graph and pharmacological feature fusion for drug-drug interaction event prediction," in *2024 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pp. 467–470, IEEE, 2024.
- [11] R. Celebi, H. Uyar, E. Yasar, O. Gumus, O. Dikenelli, and M. Dumontier, "Evaluation of knowledge graph embedding approaches for drug-drug interaction prediction in realistic settings," *BMC bioinformatics*, vol. 20, no. 1, p. 726, 2019.
- [12] M. R. Karim, M. Cochez, J. B. Jares, M. Uddin, O. Beyan, and S. Decker, "Drug-drug interaction prediction based on knowledge graph embeddings and convolutional-1stm network," in *Proceedings of the 10th ACM international conference on bioinformatics, computational biology and health informatics*, pp. 113–123, 2019.
- [13] X. Lin, Z. Quan, Z.-J. Wang, T. Ma, and X. Zeng, "Kggn: Knowledge graph neural network for drug-drug interaction prediction," in *IJCAI*, vol. 380, pp. 2739–2745, 2020.
- [14] X. Su, Z. You, D. Huang, L. Wang, L. Wong, B. Ji, and B. Zhao, "Biomedical knowledge graph embedding with capsule network for multi-label drug-drug interaction prediction," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 6, pp. 5640–5651, 2022.
- [15] X. Su, L. Hu, Z. You, P. Hu, and B. Zhao, "Attention-based knowledge graph representation learning for predicting drug-drug interactions," *Briefings in bioinformatics*, vol. 23, no. 3, 2022.
- [16] Z.-H. Ren, C.-Q. Yu, L.-P. Li, Z.-H. You, Y.-J. Guan, X.-F. Wang, and J. Pan, "Biodkg-ddi: predicting drug-drug interactions based on drug knowledge graph fusing biochemical information," *Briefings in Functional Genomics*, vol. 21, no. 3, pp. 216–229, 2022.
- [17] A. Tamir and J.-S. Yuan, "Kite-ddi: A knowledge graph integrated transformer model for accurately predicting drug-drug interaction events from drug smiles and biomedical knowledge graph," *IEEE Access*, 2025.
- [18] Y. Bai, K. Gu, Y. Sun, and W. Wang, "Bi-level graph neural networks for drug-drug interaction prediction," *arXiv preprint arXiv:2006.14002*, 2020.
- [19] Y. Wang, Y. Min, X. Chen, and J. Wu, "Multi-view graph contrastive representation learning for drug-drug interaction prediction," in *Proceedings of the web conference 2021*, pp. 2921–2933, 2021.
- [20] T. Lyu, J. Gao, L. Tian, Z. Li, P. Zhang, and J. Zhang, "Mdn: A multimodal deep neural network for predicting drug-drug interaction events," in *Ijcai*, vol. 3536, 2021.
- [21] X. Su, P. Hu, Z.-H. You, P. S. Yu, and L. Hu, "Dual-channel learning framework for drug-drug interaction prediction via relation-aware heterogeneous graph transformer," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 38, pp. 249–256, 2024.
- [22] J. Xu, S. Chen, Y. Ren, X. Shi, H. Shen, G. Niu, and X. Zhu, "Self-weighted contrastive learning among multiple views for mitigating representation degeneration," *Advances in neural information processing systems*, vol. 36, pp. 1119–1131, 2023.
- [23] P. Xiong, H. Chen, J. Zhou, Y. Zeng, and Q. Dai, "Mgrl-ddi: Multi-view graph representation learning for accurate drug-drug interaction prediction," *Journal of Chemical Information and Modeling*, 2025.
- [24] C. Chen, X. Shi, J. Nie, J. Xu, and L. Wang, "A molecular representation learning model based on multidimensional joint and cross-learning for drug-drug interaction prediction," *Journal of Chemical Information and Modeling*, vol. 65, no. 16, pp. 8889–8900, 2025.
- [25] D. Li, F. Zhao, Y. Yang, Z. Cui, P. Hu, and L. Hu, "Multi-view contrastive learning for drug-drug interaction event prediction," *IEEE Journal of Biomedical and Health Informatics*, pp. 1–12, 2025.